

COL 703/7203

Inductive step

27th August

$$\psi_1 \dots \psi_n \Rightarrow \psi$$

ψ_n can once again be an instance
of an axiom or
be present in $X \cup \{\phi\}$.

↳ The same argument as
the base case applies.

The only other possibility is that ψ_n has been derived by applying MP on ψ_j and ψ_k ($j, k < n$)

Apply MP twice

$$\begin{array}{c}
 \psi_1 \dots \psi_j \quad \quad \quad \cancel{\psi_k} \\
 \hline
 \psi_j \rightarrow \psi \\
 \hline
 x \vdash \phi \rightarrow \psi_j \quad \quad x \vdash \phi \rightarrow (\psi_j \rightarrow \psi) \\
 \hline
 x \vdash (\phi \rightarrow (\psi_j \rightarrow \psi)) \rightarrow ((\phi \rightarrow \psi_j) \rightarrow (\phi \rightarrow \psi))
 \end{array}$$

Exercise

Use deduction theorem to prove
that:

$$i) \quad \phi \rightarrow \psi, \quad \psi \rightarrow \psi_1 \vdash \phi \rightarrow \psi_1$$

$$ii) \quad \phi \rightarrow (\psi_1 \rightarrow \psi_2) \vdash \psi_1 \rightarrow (\phi \rightarrow \psi_2)$$

$$\text{iff} \quad \phi \rightarrow (\psi_1 \rightarrow \psi_2), \psi_1 \vdash \phi \rightarrow \psi_2 \quad (\text{Ded. Thm})$$

$$\text{iff} \quad \phi \rightarrow (\psi_1 \rightarrow \psi_2), \psi_1, \phi \vdash \psi_2 \quad (\text{Ded. Thm})$$

here is the derivation $\phi, \phi \rightarrow (\psi_1 \rightarrow \psi_2), \psi_1 \rightarrow \psi_2, \psi_1, \psi_2$

TODOS

$$1. \vdash (\neg\psi \rightarrow \neg\phi) \rightarrow (\phi \rightarrow \psi)$$

$$\neg\psi \rightarrow \neg\phi \quad \vdash \quad \phi \rightarrow \psi$$

$$\neg\psi \rightarrow \neg\phi, \phi \quad \vdash \quad \psi$$

$$(\neg\psi \rightarrow \neg\phi) \rightarrow ((\neg\psi \rightarrow \phi) \rightarrow \psi) \quad A3$$

MP

$$\phi \rightarrow (\neg\psi \rightarrow \phi)$$

A1

$$(\neg\psi \rightarrow \phi) \rightarrow \psi$$

MP

$$\neg\psi \rightarrow \phi$$

MP

ψ

$$2. \quad \vdash \neg\neg\phi \rightarrow \phi \qquad \neg\neg\phi \vdash \phi$$

$$(\neg\phi \rightarrow \neg\neg\phi) \rightarrow ((\neg\phi \rightarrow \neg\phi) \rightarrow \phi) \quad A3$$

$$\underline{\underline{(\neg\phi \rightarrow \neg\phi) \rightarrow ((\neg\phi \rightarrow \neg\neg\phi) \rightarrow \phi)}}$$

✓

$$\neg\neg(\neg\neg\phi \rightarrow (\neg\phi \rightarrow \neg\neg\phi)) \quad A1$$

$$\underline{\underline{\neg\neg\phi}}$$

$$(\phi_1 \rightarrow (\phi_2 \rightarrow \phi_3)) \rightarrow ((\phi_1 \rightarrow \phi_2) \rightarrow (\phi_1 \rightarrow \phi_3)) \quad (A2)$$

$$\frac{(\psi \rightarrow ((\psi \rightarrow \psi) \rightarrow \psi)) \rightarrow ((\psi \rightarrow (\psi \rightarrow \psi)) \rightarrow (\psi \rightarrow \psi))}{A1} \quad A1$$

Apply MP twice
to get $(\psi \rightarrow \psi)$

3. For all formulas ϕ ,
if ϕ is consistent then ϕ is
satisfiable.

Consistency of a set of formulas

A finite set of formulas $\{\phi_1, \dots, \phi_n\}$
consistent if the formula

$\phi_1 \wedge \phi_2 \wedge \dots \wedge \phi_n$ is
consistent.

(i.e., $\nexists \neg(\phi_1 \wedge \dots \wedge \phi_n)$)

An arbitrary set of formulas X
consistent if every finite subset
of X is consistent.

ie, if $Y \subseteq_{\text{fin}} X$ then Y is
consistent.

Maximal consistent set (MCS)

An MCS is a consistent set
that cannot be extended by adding
any formula.

X is an MCS if
for every ϕ ($\phi \notin X$)

$X \cup \{\phi\}$ is inconsistent
 $\{\phi\}$

Lemma 1. Every consistent set
can be extended into an MCS.

Lemma 2 let X be an MCS. Then

- i) $\phi \in X$ iff $\neg \phi \notin X$
- ii) $\phi \vee \psi \in X$ iff $\phi \in X$ or $\psi \in X$.

Every MCS X defines a canonical valuation v_X

where $v_X = \{ p \mapsto \text{True} \text{ iff } p \in X \}$

Then $v_X \models \phi$ iff $\phi \in X$.

Lemma 3